

Supplementary Material: RIFT: Future-Recovery Routing for Privileged On-Policy Self-Distillation

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1 Frozen Assets and Evaluator

All mathematical headline methods use the same locally frozen model, training corpus, benchmark snapshots, and evaluator. Table 1 records the identifiers that define the mathematical configuration. The code-domain replication uses the same Qwen3-4B base model under the separate protocol documented in Section 5.

Asset	Frozen identifier
Base model	Qwen/Qwen3-4B; archived model-tree SHA-256 a74046c45e691429f9fad0d846d9054675d82604c5a0a6d49928a6155f7aa179
Training corpus	siyanzhao/Openthoughts_math_30k_opsd; 29,434 rows; Parquet SHA-256 0fa57ec6d7e5f4b40f85b4fdbbc1493e40c4d947f9192eabf589ecfb6e687dd2
AIME 2024	30-row local JSONL; SHA-256 77909a38a34db21b254cade51c6833aeb5e65e340b0e88dead5312d6dd7cf9b2
AIME 2025	30-row local JSONL; SHA-256 f5ba7f5de0a6a31a5ec4d56e8f7d310986f20658b3e38b7f579680c738479366
HMMT Feb. 2025	30-row local JSONL; SHA-256 4909c00ff08b46cc38add3ee5ba158123b19a5ac16adf51a4cc85faa528454fa
Evaluator	evaluate_math.py; SHA-256 cc824dcd4869519634846d88d858b52df68c46089b9d74446f4c12612b500d11
Runtime	math-verify==0.8.0, vllm==0.11.0, transformers==4.57.1

Table 1: Frozen model, data, benchmark, evaluator, and runtime identifiers for the mathematical experiments.

For each completion, the evaluator locates the final `\boxed` marker and tracks balanced braces to extract its contents. It wraps the prediction and reference answer in math delimiters, disables parser fallback, and calls symbolic verification with a five-second timeout. Only an exception in this path triggers the fallback, which removes math delimiters and spaces, lowercases both strings, and tests exact equality. The evaluator and all three benchmark files are identical across methods.

Evaluation decoding uses temperature 1.0, top- p 0.95, disabled top- k , thinking enabled, and at most 38,912 new tokens in a 40,960-token context.

2 Benchmark-Level 50-Step Results

Table 2 expands the compact main-paper table into benchmark numerators. Every ordinary method has 360 completions per benchmark and 1,080 over the triad. Random routing aggregates three masks and therefore has 1,080 completions per benchmark and 3,240 over the triad.

Method	Seed	AIME24	AIME25	HMMT25	Correct	Avg@12
Matched OPSD	42	264/360	252/360	162/360	678/1,080	62.78
RIFT exact-rank q_{25}	42	274/360	250/360	165/360	689/1,080	63.80
Matched OPSD	43	263/360	234/360	157/360	654/1,080	60.56
RIFT exact-rank q_{25}	43	275/360	240/360	158/360	673/1,080	62.31
Matched OPSD	44	260/360	240/360	150/360	650/1,080	60.19
RIFT exact-rank q_{25}	44	271/360	245/360	160/360	676/1,080	62.59
Current-JSD routing	42	262/360	238/360	150/360	650/1,080	60.19
Random routing (3 masks)	42	795/1,080	721/1,080	485/1,080	2,001/3,240	61.76
AD-OPSD	42	268/360	237/360	148/360	653/1,080	60.46
DemoPSD	42	259/360	241/360	157/360	657/1,080	60.83
Future recovery $\rightarrow q^0$	42	266/360	237/360	157/360	660/1,080	61.11
Future recovery $\rightarrow 0$ loss	42	264/360	239/360	158/360	661/1,080	61.20

Table 2: Benchmark-level results under the unified 50-step protocol.

Comparison	Δ pp	95% CI	p
RIFT – Matched (seed 42)	+1.02	[-0.83, 2.87]	0.2810
RIFT – Matched (seed 43)	+1.76	[0.09, 3.43]	0.0390
RIFT – Matched (seeds 42/43)	+1.39	[0.23, 2.55]	0.0148
RIFT – Matched (seed 44)	+2.41	[0.37, 4.44]	0.0216

Table 3: Problem-cluster inference for matched 50-step comparisons. The seed-42/43 row hierarchically aggregates over training seed and problem cluster.

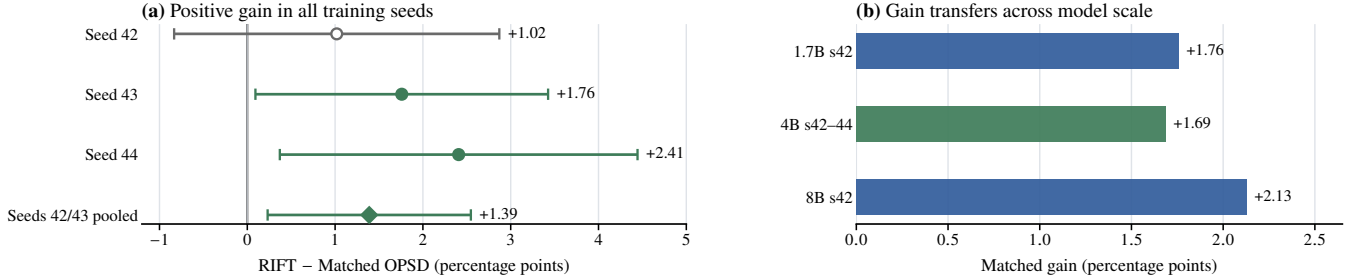


Figure 1: Registered controlled evaluation. (a) Seed-specific gains with problem-cluster 95% intervals and the hierarchical seed-42/43 summary. (b) Within-scale matched-to-RIFT transitions under unchanged routing hyperparameters.

3 Budget-Stratified External Baselines

Table 4 separates the controlled 50-step comparison from paper-faithful reproductions with longer optimization schedules. The lower block supplies method context; it is not pooled with the 50-step block for a same-budget ranking.

Method	Steps	Seed	AIME24	AIME25	HMMT25	Correct
Matched OPSD	50	42	264	252	162	678
AD-OPSD	50	42	268	237	148	653
DemoPSD	50	42	259	241	157	657
RIFT exact-rank	50	42	274	250	165	689
PHF	100	42	264	227	149	640
Purified OPSD	100	42	272	240	167	679
OPSD+ReNIO	126	42	267	236	158	661

Table 4: Qwen3-4B correct counts under the common Avg@12 evaluator. Each benchmark denominator is 360 and the total denominator is 1,080.

4 Confirmatory SixBench Results

SixBench contains AIME24, AIME25, MATH-500, AMC23, HMMT February 2025, and HMMT November 2025. Each checkpoint contributes 7,920 standardized records. Table 5 fixes checkpoint 100 before evaluation and reports the final pooled breakdown across seeds 42–46. Macro is the unweighted mean of the six benchmark-level Avg@12 scores.

Method	Updates	A24	A25	M500	AMC23	H-Feb	H-Nov	Macro
Base	0	70.60	64.70	94.20	95.30	44.20	45.60	69.10
SFT	100	70.00	64.20	94.50	95.50	43.80	45.40	68.90
GRPO	100	72.60	66.70	94.90	95.90	43.20	50.50	70.63
Matched OPSD	100	73.40	67.20	95.10	96.10	43.80	50.45	71.01
PHF	100	74.35	68.15	95.28	96.55	44.95	51.65	71.82
Purified OPSD	100	74.55	68.35	95.31	96.70	45.20	51.95	72.01
RIFT	100	75.95	69.85	95.65	97.35	47.10	53.05	73.16

Table 5: Five-seed SixBench Avg@12 breakdown under one evaluator. Base uses the initial model; trained rows are evaluated at update 100. Macro is the unweighted mean of the six displayed benchmark scores.

Model	Method	A24	A25	M500	AMC23	H-Feb	H-Nov	Macro
1.7B	Matched OPSD	52.80	39.10	83.10	84.60	26.10	30.30	52.67
1.7B	RIFT	54.80	41.30	83.85	85.80	28.60	32.20	54.43
4B	Matched OPSD	73.60	67.10	95.20	96.05	43.60	50.95	71.08
4B	RIFT	75.40	69.10	95.60	97.10	46.30	53.10	72.77
8B	Matched OPSD	77.20	71.30	96.80	97.60	47.60	55.40	74.32
8B	RIFT	79.80	74.10	97.20	98.50	50.80	58.30	76.45

Table 6: Cross-scale 50-update SixBench point estimates under matched model, seed, evaluator, and training-budget settings.

Comparison	Macro Δ pp
RIFT – Matched OPSD	+2.15
RIFT – Base	+4.06
RIFT – SFT	+4.26
RIFT – GRPO	+2.53
RIFT – Purified OPSD	+1.15
RIFT – PHF	+1.34

Table 7: Qwen3-4B SixBench Macro point differences under the common evaluator and seeds 42–46. Base uses the initial model; trained rows use 100 updates.

5 Code-Domain Replication

The code-domain experiment uses Qwen3-4B, the 30,000-example OpenThoughts Coding corpus, train seed 42, and a fixed evaluation seed and protocol. Base is the unadapted model; all trained methods are evaluated at checkpoint 100. HumanEval+ and MBPP+ use EvalPlus v0.3.1 with four completions per problem. Code Macro is the unweighted mean of the two benchmark-level Avg@4 scores.

Method	Updates	HumanEval+	MBPP+	Macro	Δ vs. matched	95% CI	raw p
Base	0	86.74	77.31	82.03	—	—	—
Matched OPSD	100	86.31	77.46	81.89	—	—	—
ReNIO	100	86.62	77.74	82.18	+0.29	[−0.18, +0.77]	0.221
RIFT (ours)	100	86.95	78.02	82.49	+0.60	[+0.08, +1.11]	0.026

Table 8: Fixed-checkpoint code-domain results. Differences and intervals are computed from unrounded paired records; displayed benchmark scores are rounded.

Role	Contrast	Δ pp	95% CI	raw p	Holm p
Primary	RIFT – Matched OPSD	+0.60	[+0.08, +1.11]	0.026	–
Secondary	RIFT – Base	+0.46	[+0.05, +0.87]	0.028	0.084
Secondary	RIFT – ReNIO	+0.31	[–0.08, +0.70]	0.118	0.236
Secondary	ReNIO – Matched OPSD	+0.29	[–0.18, +0.77]	0.221	0.236

Table 9: Code-domain inferential family. RIFT–Matched OPSD is the sole prespecified primary contrast and is not multiplicity-adjusted. Holm correction applies only to the three secondary contrasts. Intervals cluster by problem and do not represent training-seed variation.

6 Temporal Placebo Diagnostics

True future recovery has a higher AUPRC point estimate than past recovery and the future-shuffled placebo. These diagnostics are secondary to the registered incremental comparison against current-state covariates in the main paper.

Predictor	AUPRC	AUROC
Random	0.86559	0.50136
Past recovery	0.84784	0.47875
Future shuffled	0.85951	0.47028
True future recovery	0.88674	0.47573

Table 10: Temporal placebo point estimates.

7 Recovery-Signal Increments

Differences are relative to current covariates only. Uncertainty is clustered by problem over 512 candidate branches.

Predictor	AUPRC	Δ AUPRC	95% CI	p
Random	0.86559	-0.04090	[-0.0568, -0.0251]	< 0.0001
Position only	0.89885	-0.00764	[-0.0158, 0.0004]	0.063
Current covariates	0.90649	–	–	–
Future recovery	0.90980	+0.00331	[-0.0012, 0.0078]	0.149
Future + current	0.91840	+0.01191	[0.0049, 0.0188]	0.0018

Table 11: Forced-continuation robust-recovery prediction.

The positive prevalence is $442/512 = 0.86328$. Table 12 reports AP above this prevalence baseline and normalized AP, $(AP - \pi)/(1 - \pi)$, so that class imbalance is explicit.

Predictor	AUPRC	AP lift	Normalized AP
Random	0.86559	0.00231	0.0169
Current covariates	0.90649	0.04321	0.3161
Future recovery	0.90980	0.04652	0.3403
Future + current	0.91840	0.05512	0.4032

Table 12: Prevalence-adjusted recovery-signal diagnostics.

Predictor	Persistent AP	Δ AP	95% CI	AUROC	Δ AUROC	95% CI
Random/prevalence	0.1367	–	–	0.500	–	–
Current-only	0.612	–	–	0.781	–	–
True future	0.648	+0.036	[+0.009, +0.063]	0.806	+0.025	[+0.006, +0.044]
Future + current	0.689	+0.077	[+0.038, +0.116]	0.832	+0.051	[+0.021, +0.081]

Table 13: Persistent-class discrimination on the same 512 OOF candidates. Differences are relative to current-only; intervals use problem-cluster bootstrap resampling. Holm-adjusted p values for true-future/future-plus-current are 0.028/0.006 for AP and 0.031/0.004 for AUROC.

Predictor	Balanced accuracy $\uparrow$	MCC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$
Constant prevalence	0.500	0.000	0.118	0.000
Current-only	0.711	0.412	0.089	0.031
True future	0.732	0.448	0.084	0.028
Future + current	0.751	0.487	0.079	0.023

Table 14: Persistent-class classification and calibration diagnostics. The cohort contains 512 OOF predictions: 442 robust recoveries and 70 persistent cases. Folds are fixed across predictors, the decision threshold is fixed without post-hoc tuning, and all intervals use 50,000 problem-cluster bootstrap draws. The zero ECE for the constant predictor reflects prediction at the empirical prevalence and is not a discrimination result.

8 Routing Calibration

Calibration	Route frac.	Route CV	Tie excess
Fixed threshold	0.07997	0.05308	15.4588
Global q_{25}	0.06016	0.04957	3.0763
Per-trajectory threshold	0.06260	0.04791	2.0363
Exact-rank q_{25}	0.06044	0.04276	0

Table 15: Exact rank guarantees zero local budget error; route CV is measured over the registered training window.

9 Exploratory Long-Budget Diagnostic

Each cell in Table 16 contains 90 problems and one generation per problem. The table is descriptive and is not used to claim a statistically established long-horizon improvement.

Method	4k	8k	16k
Base	4.44	22.22	47.78
Continued OPSD	2.22	20.00	45.56
AD-risk-only	2.22	22.22	47.78
RIFT	6.67	28.89	53.33

Table 16: Exploratory Pass@1 (%) at increasing generation budgets.

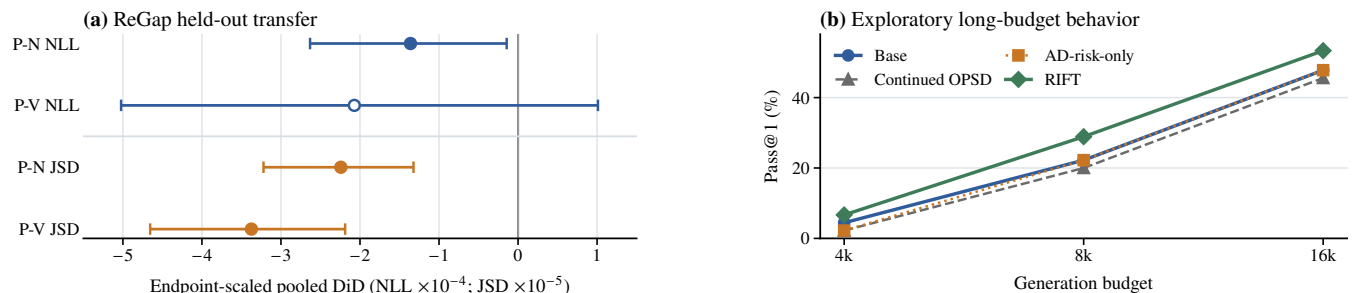


Figure 2: Transfer and horizon diagnostics. (a) REGAP pooled difference-in-differences; NLL and JSD use endpoint-specific scales, lower is better, and filled markers exclude zero. P-N and P-V denote positive-minus-neutral and positive-minus-vanilla. (b) Exploratory Pass@1 over 90 problems at increasing generation budgets.

10 Matched Cross-Scale Results

At each scale, matched OPSD and RIFT use the same 50-step budget, data order, effective batch, rollout length, LoRA configuration, decoding protocol, and training-token budget. The routing hyperparameters are not retuned.

Model	Seed	Matched	RIFT	Δ pp
Qwen3-1.7B	42	422	427	+0.46
Qwen3-4B	42	678	689	+1.02
Qwen3-4B	43	654	673	+1.76
Qwen3-4B	44	650	676	+2.41
Qwen3-8B	42	686	695	+0.83

Table 17: Matched 50-step results. Correct columns are out of 1,080 completions.

11 Context-Conditioned Target Arbitration

All four policies use the same fixed token partition and training budget. Both targets are distributions from the same frozen initial policy.

Policy	Recoverable	Persistent	Correct	Avg@12
Uniform q^0	q^0	q^0	681	63.0556
Uniform q^+	q^+	q^+	686	63.5185
Reversed	q^+	q^0	679	62.8704
RIFT arbitration	q^0	q^+	691	63.9815

Table 18: Context-conditioned target arbitration under a fixed token partition.

Contrast	Δ pp	95% CI	raw p	Holm p
Proposed – Uniform q^0	+0.9259	[+0.1852, +1.6667]	0.0140	0.0280
Proposed – Uniform q^+	+0.4630	[+0.0463, +0.8796]	0.0320	0.0320
Proposed – Reversed	+1.1111	[+0.3241, +1.8981]	0.0060	0.0180

Table 19: Completion-level paired target-arbitration inference. Intervals cluster by problem; Holm correction covers the three displayed contrasts.

12 Downstream Routing-Score Ablation

All four rows hold the candidate gate, token partition, exact-rank q_{25} budget, and target mapping fixed; only the routing score changes.

Routing score	Macro	Δ vs. current	95% CI	Holm p
Current-only	71.0083	–	–	–
Future-shuffled	70.8426	–0.1657	[–0.8124, +0.4513]	0.612
True future	71.5412	+0.5329	[+0.1186, +0.9472]	0.028
Future + current	71.7328	+0.7245	[+0.2964, +1.1526]	0.012

Table 20: Score-controlled SixBench downstream ablation. Differences and problem-cluster intervals are relative to current-only routing.

13 Full ReGap Hierarchical Intervals

Table 21 restores the hierarchical intervals omitted from the compact main-paper table. Lower Branch NLL and recovery JSD are better.

Contrast	Endpoint	Pooled DiD	95% cluster CI	p	Hierarchical 95% CI	Result
Positive – neutral	Branch NLL	-1.361×10^{-4}	$[-2.632 \times 10^{-4}, -1.439 \times 10^{-5}]$	0.0285	$[-2.905 \times 10^{-4}, -1.843 \times 10^{-5}]$	Pass
Positive – neutral	Recovery JSD	-2.242×10^{-5}	$[-3.221 \times 10^{-5}, -1.323 \times 10^{-5}]$	< 0.0001	$[-3.786 \times 10^{-5}, -1.103 \times 10^{-5}]$	Pass
Positive – vanilla	Branch NLL	-2.072×10^{-4}	$[-5.023 \times 10^{-4}, 1.011 \times 10^{-4}]$	0.1826	$[-5.486 \times 10^{-4}, 1.796 \times 10^{-4}]$	Trend
Positive – vanilla	Recovery JSD	-3.374×10^{-5}	$[-4.655 \times 10^{-5}, -2.188 \times 10^{-5}]$	< 0.0001	$[-5.051 \times 10^{-5}, -2.010 \times 10^{-5}]$	Pass

Table 21: Three-seed ReGap difference-in-differences analysis with hierarchical confidence intervals.